each with a yellow-edged, bluish, brown or purple lip, are produced in spring. Has ovate or lance-shaped leaves, 3–5in (8–12cm) long. Cultivate as for *O. fuciflora* of gardens. Z8–10
O. holoserica of gardens. See *O. fuciflora*.
O. lutea (illus. p.467). Deciduous, terrestrial orchid. H 3–12in (8–30cm). In spring, bears short spikes of flowers, ½in (1cm) long, with greenish sepals, yellow petals, and brown-centered, bright yellow lips. Has ovate, basal leaves, 2–4in (5–10cm) long. Cultivate as for *O. fuciflora*. Z8–10
O. speculum, syn. *O. vernixia* (Mirror orchid). Deciduous, terrestrial orchid. H 3–12in (8–30cm). In spring, bears dense spikes of flowers, ½in (1cm) long, with greenish or yellow sepals, purple petals, and 3-centered, brown lips. Has oblong to lance-shaped leaves, 1½–3in (4–7cm) long. Cultivate as for *O. fuciflora*. Z8–10
O. sphegodes, syn. *O. aranifera* (Early spider orchid). Deciduous, terrestrial orchid. H 4–18in (10–45cm). In spring-summer, bears spikes of flowers, ½in (1cm) long, that vary from green to yellow and have spiderlike, blackish-brown marks on lips. Leaves are ovate to lance-shaped and 1½–3in (4–8cm) long. Cultivate as for *O. fuciflora*. Z8–10
O. tenthredinifera (Sawfly orchid; illus. p.466). Deciduous, terrestrial orchid. H 6–22in (15–55cm). In spring, bears spikes of flowers, ½in (1cm) long, in colors of white to pink, or blue and green, each with a violet or bluish lip edged with pale green. Has a basal rosette of ovate to oblong leaves, 2–3½in (5–9cm) long. Cultivate as for *O. fuciflora*. Z9–11
O. vernixia. See *O. speculum*.

Ophthalmophyllum. See *Conophytum*.

OPLISMENUS

POACEAE/GRAMINEAE

See also GRASSES, BAMBOOS, RUSHES, and SEDGES.
O. africanus. See *O. hirtellus*.
'Variegatus' see *O.h.* 'Variegatus'.
O. hirtellus, syn. *O. africanus*. Evergreen perennial. H 6in (15cm), but may form mounds to 3ft (90cm), S indefinite. Has wiry stems, spreading and rooting at the nodes, bearing narrowly lance-shaped to ovate, softly hairy, mid-green leaves, with long points. Small flowers are produced in one-sided racemes, to 6in (15cm) long, from summer to winter. Z10–11 (min. 41°F/5°C). **'Variegatus'** (syn. *O. africanus* 'Variegatus') illus. p.469.

OPUNTIA

Prickly pear

CACTACEAE

Genus of perennial cacti, ranging from small, alpine, groundcover plants to large, evergreen, tropical trees, with at times insignificant glochids—short, soft, barbed spines produced on areoles. Mature plants produce abundant, short-spined, pear-shaped, green, yellow, red, or purple fruits (prickly pears), edible in some species. Borderline fully hardy to frost tender. Hardy species must be kept dry in winter in order to survive low temperatures. Needs sun and well-drained soil. Water containerized specimens when in full growth. Propagate by seed or root stem segments in spring or summer. Some species can become invasive. Contact with the bristles causes intense skin irritation and they are difficult to remove.
O. brasiliensis. See *Brasiliopuntia brasiliensis*.
O. erinacea* var. *ursina. See *O. polyacantha* var. *erinacea*.
O. ficus-indica (Indian fig). Bushy to treelike, perennial cactus. H and S 15ft (5m). Has flattened, oblong, blue-green stem segments, 20in (50cm) long, and spineless. In spring–summer, produces shallowly saucer-shaped, yellow flowers, 4in (10cm) across, followed by edible, purple fruits. Z10–11 (min. 37–41°F/3–5°C)
O. humifusa illus. p.494.
O. microdasys (Bunny ears). Bushy, perennial cactus. H and S 60cm (2ft). Has flattened, ovate, green stem segments, 3–7in (8–18cm) long, that develop brown marks in low temperatures. Produces spineless areoles, with white, yellow, brown, or red glochids, closely set in diagonal rows. Abundant, funnel-shaped, yellow flowers, 2in (5cm) across, are borne in summer on plants over 6in (15cm) tall, followed by small, dark red fruits. Z10–11 (min. 37–41°F/3–5°C). **var. *albispina*** illus. p.483.
O. polycantha* var. *erinacea, syn. *O. erinacea* var. *ursina*, illus. p.481.
O. robusta illus. p.488.
O. tunicata. See *Cylindropuntia tunicata*.

ORBEA

ASCLEPIADACEAE/APOCYNACEAE

Genus of clump-forming, perennial succulents with erect, 4-angled stems. Stem edges are often indented and may produce small leaves that drop after only a few weeks. Frost tender. Needs sun or partial shade and well-drained soil. Propagate by seed or stem cuttings in spring or summer.
O. variegata, syn. *Stapelia variegata* (Star flower, Toad cactus), illus. p.488.

ORCHIDS

ORCHIDACEAE

Family of perennials, some of which are evergreen or semievergreen, grown for their beautiful, unusual flowers. These consist of 3 outer sepals and 3 inner petals, the lowest of which, known as the lip, is usually enlarged and different from the others in shape, markings, and color. There are about 750 genera and 22,500 species, together with an even greater number of hybrids, bred partly for their vigor and ease of care. They are divided into epiphytic and terrestrial plants. (The spread of an orchid is indefinite.)

Epiphytic orchids
Epiphytes have more flamboyant flowers than terrestrial orchids and are more commonly grown. In the wild, they grow on tree branches or rocks (lithophytes), obtaining nourishment through clinging roots and moisture through aerial roots. Most consist of a horizontal rhizome, from which arise vertical, water-storing, often swollen stems known as pseudobulbs. Flowers and foliage are produced from the newest pseudobulbs. Other epiphytes consist of a continuously growing upright rhizome; on these, flower spikes are produced in the axils of leaves growing from the rhizome. In temperate climates, epiphytes need to be grown under glass.

Cultivation of epiphytes
For cultivation purposes, epiphytes, which are all frost tender, may be divided into 3 groups: cool-greenhouse types, which require min. 50°F (10°C) and max. 75°F (24°C); intermediate-greenhouse types, needing 55–80°F (13–27°C); and warm-greenhouse types, requiring 65–80°F (18–27°C). Summer temperatures need to be controlled by ventilation and shading the glass. Cool-greenhouse orchids may be placed outdoors in summer; this improves flowering. Other types may also be grown outdoors if the air temperature remains within these ranges.
The amount of light required in summer is given in individual plant entries. All epiphytic orchids, however, need to be kept out of direct sun in summer to avoid scorching, and require full light in winter.
Epiphytic orchids, whether grown indoors or outside, need a special soil-free compost obtained from an orchid nursery or made by mixing 2 parts fibrous material (such as barchippings and/or peat) with 1 part porous material (such as moss and/or expanded clay pellets). Most epiphytes may be grown in pots, but some may be successfully cultivated in a hanging basket or on a slab of bark (with moss around their roots) suspended in the greenhouse.
In summer, water plants freely and spray regularly. Those suspended on bark slabs need a constantly moist atmosphere. In winter, water moderately; if plants are in growth, spray occasionally. Some orchids rest in winter, when they need scarcely any water or none at all. Orchids benefit from weak foliar feeds; apply as for watering. Repot every other year, in spring; if plants are about to flower, repot after flowering.

Terrestrial orchids
Terrestrial orchids, some of which also produce pseudobulbs, grow in soil or leaf mold, sustaining themselves in the normal way through roots or tubers. Some may be grown in borders, but many in temperate climates need to be grown in pots and protected under glass in winter.

Cultivation of terrestrial orchids
Terrestrial orchids are fully hardy to frost tender. *Cypripedium* species may be grown outdoors in any area, preferably in neutral to acidic soil, but cannot withstand severe frost, if frozen solid in pots or without snow cover, or tolerate very wet soil in winter. Other terrestrial orchids, except in very mild areas, are best grown in pots; use the same compost as for epiphytes but add 1 part grit to 2 parts compost. Place pots outdoors in a peat bed or in a glasshouse in the growing season. Keep dry when dormant. Under glass, light requirements, watering, feeding, and repotting are as for epiphytes.

Orchid propagation
Any orchids with pseudobulbs (for example, *Cattleya*, *Cymbidium*, *Dendrobium* and *Oncidium*), may be increased by removing and replanting old, leafless pseudobulbs when repotting in spring. Take care to retain at least 4 pseudobulbs on the parent plant. Some orchids without pseudobulbs produce new growth from the base. When a plant has 6 new growths, divide it in spring into 2 and repot both portions. Propagate *Disa*, *Paphiopedilum* and *Phragmipedium* in this way. Large specimens of *Eria* and *Masdevallia* may be divided in spring, leaving 4–6 stems on each portion.
Propagation of *Phalaenopsis* is by stem cuttings taken soon after flowering. *Vanda* may be increased by removing the top half of the stem once it has produced aerial roots and leaves; new growths will develop from the leafless base. Achieving success with these methods is difficult and not recommended for the beginner.
Propagate terrestrial orchids with tubers by division of the tubers. Genera that may be increased in this way are: *Cypripedium* (in spring), *Dactylorhiza* (spring), *Ophrys* (fall), *Serapias* (fall), and *Spiranthes* (spring). *Angraecum* should not be propagated in cultivation, because the parent plant is easily endangered.

Orchids are illustrated on pp.466–47.
See also x *Aliceara*, *Angraecum*, *Anguloa*, *Bletilla*, *Brassavola*, *Bulbophyllum*, *Calanthe*, *Calypso*, *Cattleya*, x *Cattlianthe*, *Coelogyne*, *Cymbidium*, *Cypripedium*, *Dactylorhiza*, *Dendrobium*, *Dendrochilum*, *Epidendrum*, *Eria*, *Gomesa*, *Guarianthe*, *Laelia*, *Lycaste*, *Masdevallia*, *Maxillaria*, *Miltonia*, *Miltoniopsis*, *Oncidium*, x *Oncidopsis*, *Ophrys*, *Paphiopedilum*, *Phaius*, *Phalaenopsis*, *Phragmipedium*, *Pleione*, *Psychopsis*, x *Rhyncattleanthe*, *Rossioglossum*, *Serapias*, *Spiranthes*, *Stanhopea*, *Vanda* and *Zygopetalum*.

Orchis elata. See *Dactylorhiza elata*.
Orchis maderensis. See *Dactylorhiza foliosa*.
Orchis morio. See *Anacamptis morio*.

OREOCEREUS

CACTACEAE

Genus of mainly columnar, perennial cacti with thick, cylindrical, much-ribbed stems, usually branching from the base, with spines and, in some species, covered in long hairs. Solitary, tubular-funnel-shaped flowers are produced near the stem tips during the day in summer. Frost tender. Requires full sun and very well-drained, slightly alkaline soil. Water well during spring and summer, much less so during fall and winter. Propagate by seed in spring or summer.
O. aurantiacus. See *Matucana aurantiaca*.
O. celsianus, syn. *Cleistocactus celsianus* (Old man of the Andes), illus. p.493.
O. trollii, syn. *Cleistocactus trollii* (Old man of the Andes). Slow-growing, columnar, perennial cactus. H 28in (70cm), S 4in (10cm). Cylindrical, green stem, 3–4in (7–10cm) wide, with thick, golden spines is almost hidden by long, wispy, hairlike, white spines. Bears pink flowers, recurved